Rohan Batra

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SUMMARY

Final-year B.Tech Computer Science (Cyber Security) student at UPES (graduating 2027, CGPA 8.1) open to any role at HPE. Co-built and operates a LAN-first campus emergency communication system on HPE Juniper SRX, EX and Mist hardware — presented to Team HPE in Bengaluru — and maintains it hands-on: configuring and updating the hardware and software, troubleshooting incidents to root cause, and running health checks and load tests. Programs in Python and C++; comfortable across Linux, containers, cloud and the networking beneath them.

EDUCATION

UPES Dehradun — B.Tech (Hons.) Computer Science Engineering, Cyber Security *Aug 2023 – Aug 2027*
CGPA 8.1/10

SKILLS

Languages: Python, C++, C, Rust, TypeScript, JavaScript, Shell
Networking: HPE Juniper SRX / EX / Mist, Junos, Virtual Chassis, VLANs, QoS / CoS, LLDP-MED, PoE, NETCONF, DHCP / NAT, DNS, Subnetting, pfSense
Infrastructure: Linux, Kubernetes, Docker, Helm, Nginx, Proxmox, QEMU, Terraform / OpenTofu
DevOps: GitHub Actions, CI gating (Ruff, ShellCheck, gitleaks), cargo-llvm-cov, cargo-mutants
Security: Gitleaks, HashiCorp Vault, Semgrep, Trivy, age encryption, AppArmor, Burp Suite, Cerbos
Data: MongoDB, Redis, PostgreSQL, ChromaDB, MySQL, DuckDB, DuckLake
Backend: FastAPI, Flask, Pydantic, SQLAlchemy, Alembic, Microservices, REST APIs

PROJECTS

UPES-ECS — Campus Emergency Communication System *Live* / *Code*
LAN-only emergency calling on HPE Juniper networking. Presented to Team HPE at HPE's Bengaluru campus (15 Jul 2026) and demonstrated live at UPES (20 Jul 2026).
Juniper SRX/EX/Mist, Junos, Asterisk/PJSIP, FastAPI, NETCONF, PowerShell

- Co-built a **LAN-only** campus emergency calling system (one-number 111 hotline) on Asterisk/PJSIP requiring no internet, SIM, or cloud — designed, deployed and operated end-to-end on an **HPE Juniper** networking backbone.
- Configured the enterprise fabric: **Juniper SRX300/320** as the guarded edge (firewall, DHCP, NTP, LAN-only boundary), a **Virtual Chassis of two EX2300-C-12P** PoE switches carrying the voice network, and two **Mist AP32** Wi-Fi 6 access points for campus-wide softphone reach.
- Engineered the voice fabric for zero-touch operation and self-healing: **LLDP-MED** auto-provisioning of voice VLAN, PoE class and DSCP; strict-priority **CoS for RTP**; **DHCP-snooping with Dynamic ARP Inspection** to protect the failover path; and Junos **event-options** policies that power-cycle a hung AP without human intervention. Budgeted PoE against the switch's real 124 W ceiling.
- Designed **offline caller location by switch port** (IP to MAC to port to room) so the network identifies where a 111 call originated with no cloud or location service, and fed live switch/AP state to the Operations Console over **NETCONF**.
- Operated it as a production system where downtime is a safety defect — active/standby high availability, health checks, load testing, and incident root-cause analysis. Three documented RCAs: a serialized single-threaded proxy, a NAT media-address fault dropping every call at 32 seconds, and leaked channels with no teardown path.
- Developed a FastAPI status/control API exposing real-time queue, call, and system state to an operations console and live LED-TV wallboards, provisioned end-to-end via a single PowerShell installer.
- Implemented multilingual IVR across **44 languages** with Piper TTS voice prompts and an offline panic-coach fallback, gated by CI (Ruff, ShellCheck, PSScriptAnalyzer, gitleaks).

Shimpaku — Juniper Junos Lab as Code *Code*
Hands-on Juniper by HPE sprint across routing, switching, security, data centre and automation.
vJunos, libvirt/KVM, OSPF, BGP EVPN-VXLAN, BFD, PyEZ/NETCONF, Ansible

- Built a **topology-as-code** vJunos lab on plain libvirt/KVM: per-link bridges, isolated management with nftables, qcow2 overlays for instant snapshot/rollback on every "break it" drill.
- Built a **spine/leaf EVPN-VXLAN** fabric: OSPF underlay, iBGP EVPN overlay, ECMP; found that a dead spine took the 40 s OSPF dead interval to detect, and closed it with **BFD** (300 ms × 3) as an 8-line generated diff.
- Automated the fabric end to end: `gen_fabric.py` renders Junos config from the topology, `deploy.py` commits over **NETCONF (PyEZ)** or Ansible and asserts OSPF, ECMP, EVPN and cross-VXLAN reachability — idempotent, drift-repairing, and checked in GitHub Actions CI.

Paridyum — Independent Multi-Tenant Platform *Live*
Self-directed systems project: built solo from an empty directory to a running five-node Kubernetes cluster on self-hosted metal with a public internet edge.
Rust, OpenTofu, Ansible, Talos Linux, Kubernetes, Cilium, PostgreSQL

- Built a **specification-as-software governance system**: 251 versioned artifacts (constitution, architecture decision records, standards, RFCs) with machine checks mapped outward to **NIST SSDF, CIS v8.1 and NIST CSF 2.0**, a multi-axis test-type taxonomy, and a precedence order in which existing code is the lowest authority.
- Delivered the full infrastructure path: hardened bare metal (AppArmor, nftables, confined QEMU/libvirt) through **Proxmox and OpenTofu** to **Talos Linux and Kubernetes** with Cilium as CNI and kube-proxy replacement, L2-announced load-balancer VIPs, and etcd snapshots that assert they captured something.
- Wrote **31,000+ lines of Rust** across two production tools — hexagonal architecture, trait-based abstraction, injectable clock adapters, and three independent SHA-256 hash chains for tamper-evident audit records.

AutoTriage — Autonomous Vulnerability Triage & Remediation Agent *Code*
An agent that triages real scanner findings end-to-end, with evals and human-in-the-loop guardrails.
Python, Claude Agent SDK, Semgrep, Trivy, Gitleaks, Pydantic

- Built an autonomous vulnerability-triage agent on the Claude Agent SDK that ingests Semgrep, Trivy, and Gitleaks findings, reasons about severity and business impact, and files tickets, assigns owners, and drafts remediation PRs.
- Enforced human-in-the-loop guardrails via Pydantic validators — confidence-threshold escalation, fail-closed on malformed output, and prompt-injection defenses that treat scanner output as untrusted input.
- Built a fix-validation loop that applies proposed patches to an isolated copy of the code and re-runs the scanner, accepting a fix only when the finding is verifiably resolved with no regression.
- Designed an eval harness scoring triage decisions against a labelled ground-truth set: **100% verdict accuracy** across 44 live findings triaged into 24 tickets and 20 human escalations, backed by 68 tests at 89% coverage.

EXPERIENCE

Hardware & WireGuard Research Intern *Jul 2026 – Present*
Exaflair Technologies Pvt. Ltd., Dehradun — Remote

- Primary infrastructure contributor to **DEKLO**, Exaflair's self-hosted edge PaaS: Ansible-automated **k3s** fleet over a **WireGuard** mesh — a Mumbai hub plus edge nodes on consumer broadband in Chandigarh — with node hardening, fleet conventions, and CloudNativePG PostgreSQL.
- Built backups for every tenant database with restore drills, turned on container-registry authentication, and surfaced node-install failures and live node state in the provisioning portal.
- Wrote the product-research site behind the platform design: instrumented trials, recorded decisions, and measured cross-carrier latency and packet loss.

Platform Engineering Intern *Nov 2025 – Jul 2026*
Eve (TechyMonks), Bengaluru — Remote

- Founded the **Eve Gateway Stack** on APISIX and etcd — central routing, JWT authentication, rate limiting, CORS, and Prometheus metrics — establishing the platform's central trust and routing layer, with gateway-level tenant isolation, trusted tenant/user header injection, and Cerbos-based authorization rules.
- Founded and led the **Eve Data Management Service**: a FastAPI microservice with source/destination/connection APIs, SQLModel models, Alembic migrations, an Airbyte Config API client, sync orchestration, webhook routes, health checks, and Kubernetes/EKS deployment.
- Built **Airbyte-based ingestion** — typed connector definitions, USER vs PLATFORM destination modes, connection testing, sync polling and history tracking, and S3/MinIO bucket auto-creation — and resolved staging failures across Airbyte URLs, timeouts, secrets, readiness probes, and Temporal support.

Technical Head *Sep 2025 – Present*
Google Developer Groups On Campus (GDGOC), UPES Dehradun

- Led and delivered technical workshops for **100+ participants** across cloud, system design, and full-stack development.

Linux Support & Community Volunteer *Sep 2023 – Apr 2025*
LinuxWale, Dehradun

- Onboarded **10+ students** to Linux — distribution selection, dual-boot setup, and driver/hardware troubleshooting across varied hardware.
- Wrote shell scripts to automate post-install configuration and routine setup tasks.

CERTIFICATIONS

- **Open Learning — Migrating from the Cisco CCNA to the JNCIA-Junos**, HPE Networking Education Services (Sep 2026)
- **In progress, via the Shimpaku Juniper sprint:** JNCIA-Junos, JNCIA-Cloud, JNCIA-SEC, JNCIA-MistAI, JNCIA-Design, JNCIA-DevOps, JNCIA-DC

ACHIEVEMENTS

- **Finalist** — HACKIITK 2026, IIT Kanpur: top 0.4% of 1,300+ teams (9,000+ participants), with an offline AI policy-gap analyser (Code)
- **Top 15%** — TryHackMe cybersecurity learning platform
- **Semifinalist** — Eureka! Junior 2022–23, E-Cell IIT Bombay (2022)
- **Semifinalist** — Eureka! Junior 2021–22, E-Cell IIT Bombay (2021)